E-COMMERCE SALES AND PROFITABILITY WAREHOUSE

SQL-Based Data Modeling and KPI Analysis of Business Performance

Submitted By:

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Tools Used:

MySQL

Key Areas:

Star Schema Design

KPI Analysis (AOV, LTV, Repeat Rate, Return Rate)

Monthly Trend Analysis

Project Type: Independent Project

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1. EXECUTIVE SUMMARY

This project presents the end-to-end development of a relational database system and its analytical application using SQL on a real-world e-commerce dataset. The primary aim of the project was to transform raw, unstructured transactional data into a well-organized relational database and extract meaningful business insights.

The dataset comprises multiple interconnected tables capturing key business entities such as customers, orders, sellers, products, payments, and delivery information. A normalized database schema was designed to ensure data consistency, minimize redundancy, and enable efficient querying. An Extended Entity Relationship (EER) diagram was developed to visually represent the structure and relationships within the database.

The analytical component of the project involved writing SQL queries ranging from basic aggregations to advanced techniques such as subqueries, Common Table Expressions (CTEs), and window functions. These were used to analyse sales performance, customer behaviour, seller efficiency, and delivery operations.

Key findings from the analysis indicate that revenue is concentrated among a limited number of sellers, customer purchasing behaviour shows repeat patterns, and delivery inefficiencies exist in certain regions. Based on these insights, practical business recommendations were formulated.

Overall, the project demonstrates strong technical proficiency in SQL, database design, and analytical thinking, making it relevant for both academic and professional applications.

2. INTRODUCTION

Background

In today's digital economy, e-commerce platforms generate large volumes of structured data through customer transactions, product listings, and logistics operations. Efficient storage and analysis of this data are essential for understanding business performance and improving decision-making processes.

Relational database systems provide a structured way to manage such data, while SQL enables efficient querying and analysis. By leveraging these tools, businesses can uncover patterns related to sales trends, customer preferences, and operational bottlenecks.

Problem Statement

Despite the availability of large datasets, extracting actionable insights is not straightforward. Raw data is often fragmented, inconsistent, and lacks immediate interpretability. Without a well-designed database and analytical framework, businesses cannot effectively utilize their data.

This project addresses the challenge of transforming raw e-commerce data into a structured relational database and performing analysis to uncover meaningful insights.

Scope of the Project

This project focuses on the following areas-

- Designing a relational database schema
- Creating an EER diagram to illustrate relationship between different tables
- Cleaning and revamping raw data
- Performing SQL based analysis across various business domains

The scope of this study is limited to descriptive and diagnostic analysis, and does not include predictive modelling.

3. PROJECT OBJECTIVES

The primary objective of this project is to demonstrate the practical application of SQL in designing and analysing a relational database using real-world data. The project is structured to achieve both technical and analytical goals.

Primary Objectives

- To design a normalized relational database consisting of multiple interconnected tables
- To establish relationships between entities using primary and foreign keys
- To ensure data integrity and minimize redundancy through proper schema design
- To clean and preprocess raw data to make it suitable for analysis

Analytical Objectives

- To analyse overall sales performance and identify revenue trends over time
- To study customer purchasing behaviour, including frequency and regional distribution
- To evaluate seller performance and identify top contributors to revenue
- To analyse delivery timelines and identify inefficiencies in logistics operations
- To derive cross-functional insights by combining multiple datasets

Technical Objectives

- To implement SQL queries using joins, aggregations, subqueries, and window functions
- To optimize queries for better performance and readability
- To present findings using visualizations for better interpretability

The project aims to bridge the gap between raw data and actionable insights, demonstrating how SQL can be used as a powerful tool for business intelligence.

4. DATASET DESCRIPTION

Data Source

The dataset used is a Brazilian e-commerce Public dataset (Olist), obtained from Kaggle containing transactional and operational data. The dataset is originally in Portuguese, therefore certain fields, such as product category names, were translated into English using a category translation table to improve readability and analysis.

It includes multiple tables representing different entities, allowing for relational database design and multi-dimensional analysis.

Key Components of the Dataset

- Customers: Contains customer IDs and geographic information
- Orders: Includes order IDs, timestamps, and status
- Order Items: Details of products within each order
- Payments: Information on payment values and methods
- Products: Product categories and attributes
- Sellers: Seller details and locations
- Reviews: Customer feedback and ratings
- Geolocation: Regional data for mapping analysis
- Category Translation: Maps product categories in English

Characteristics of the Dataset

- Highly relational, enabling complex joins
- Time-based data allowing trend analysis
- Multi-entity structure supporting cross-domain insights

Why This Dataset is Suitable

- Reflects real-world business scenarios
- Allows analysis across multiple dimensions
- Provides opportunities for both basic and advanced SQL operations

The dataset enables a holistic analysis of the e-commerce ecosystem, making it ideal for demonstrating database design and analytical capabilities.

5. DATABASE DESIGN

The database for this project was designed using a **relational model**, ensuring that data is structured efficiently and relationships between entities are clearly defined. The design process focused on creating a scalable and normalized schema that supports complex analytical queries while maintaining data integrity.

5.1 Schema Overview

The overall schema consists of multiple interconnected tables, each representing a distinct entity within the e-commerce system. The schema was designed following normalization principles to reduce redundancy and ensure consistency.

Key Features of the Schema

- Separation of data into logical entities (customers, orders, products, etc.)
- Use of **primary keys** to uniquely identify each record
- Use of **foreign keys** to establish relationships between tables
- Structured design enabling efficient joins and aggregations

5.2 Table Structure

Each table in the database was carefully designed to represent a specific component of the business. The structure ensures clarity, scalability, and ease of querying.

Main Tables and Their Roles

- Orders: Acts as the central transactional table, connecting multiple entities and enabling end-to-end analysis of the order lifecycle
- Order Items: Serves as a bridge between orders, products, and sellers, enabling item-level analysis within each transaction
- Customers and Sellers: Represent the primary actors in the system and are linked to transactions through unique identifiers
- Products: Provides product-level classification, enabling category-based analysis
- Payments and Reviews: Extend the transactional data by adding financial and customer feedback dimensions

- Geolocation and Category Translation: Enhance the dataset by adding spatial context and improving interpretability

5.3 Relationships Between Tables

Relationships between tables were established using foreign keys to maintain referential integrity and ensure accurate data linkage.

- **Customers → Orders**
    - One-to-Many relationship
    - A customer can place multiple orders
- **Orders → Order Items**
    - One-to-Many relationship
    - Each order can contain multiple products
- **Orders → Payments**
    - One-to-One / One-to-Many (depending on dataset)
    - Each order has associated payment information
- **Orders → Reviews**
    - One-to-One relationship
    - Each order may have a review
- **Order Items → Products**
    - Many-to-One relationship
    - Each item corresponds to a product
- **Order Items → Sellers**
    - Many-to-One relationship
    - Each item is fulfilled by a seller

Importance of Relationships

- Enable multi-table joins for comprehensive analysis
- Maintain consistency and accuracy of data
- Support cross-functional insights such as linking customer behaviour with sales and delivery performance

5.4 Normalization and Design Efficiency

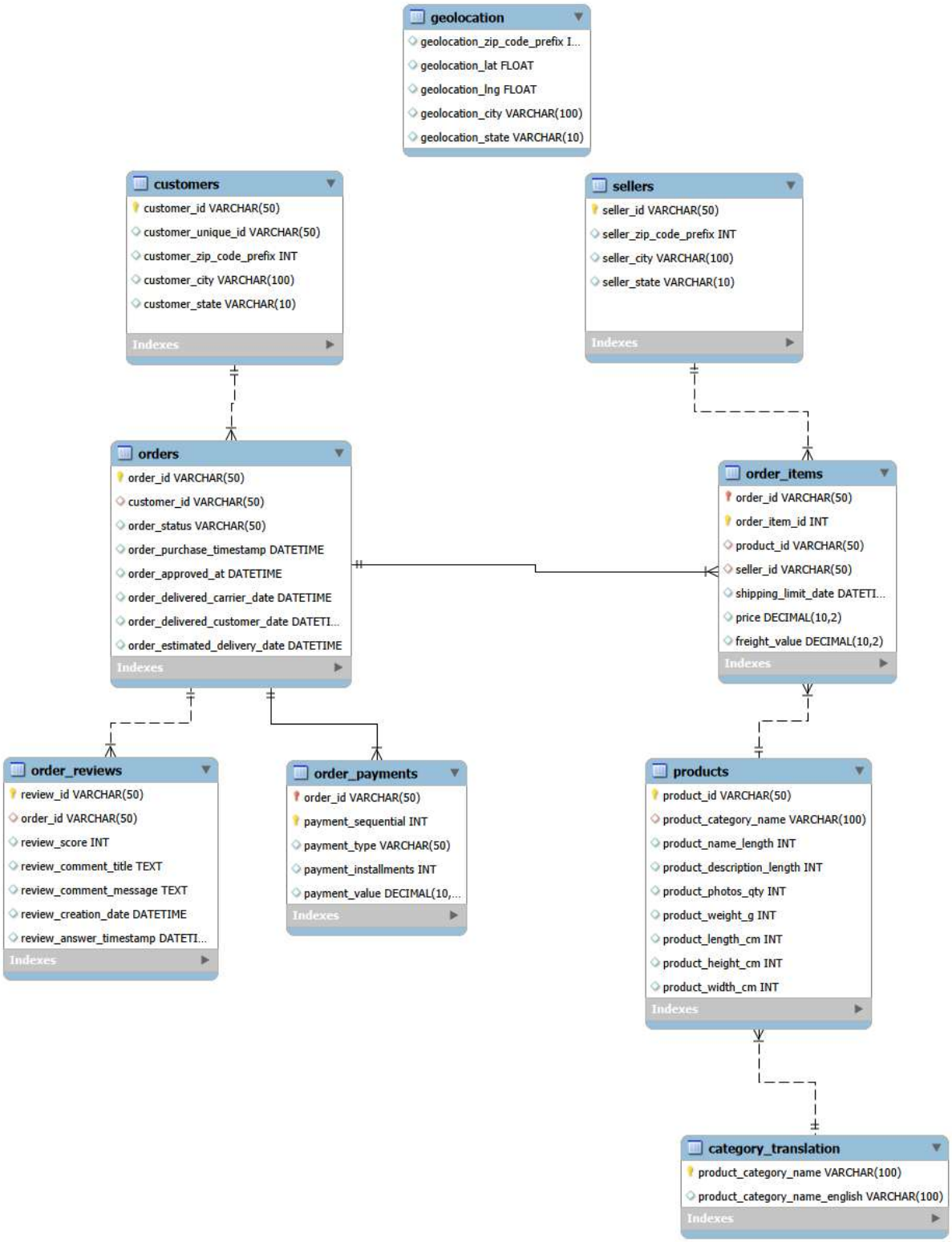
The database is designed following normalization principles to minimize redundancy and ensure efficient data management. Each entity is stored separately, and relationships are maintained using foreign keys.

Benefits of the Design

- Reduces data duplication
- Improves data consistency and integrity
- Simplifies updates and maintenance
- Enhances query performance

The normalized structure ensures that the database remains scalable and capable of handling complex analytical queries without compromising performance.

6. ENTITY RELATIONSHIP DIAGRAM (ERD)



The Extended Entity Relationship (EER) diagram represents the logical structure of the database and illustrates how different entities within the e-commerce system are interconnected. The database consists of multiple tables including customers, orders, order items, products, sellers, payments, reviews, geolocation, and category translation.

A key feature of the design is the **central role of the orders table**, which connects customers, payments, reviews, and order items. This structure enables efficient tracking of the entire order lifecycle and supports complex analytical queries.

The relationships in the database are primarily **one-to-many**, such as customers to orders and orders to order items. The **order items table acts as a bridge**, resolving the many-to-many relationship between orders and products, while also linking sellers to each transaction. This is a critical component that enables product-level and seller-level analysis.

Additional supporting tables such as geolocation and category translation enhance the dataset by providing regional context and improving interpretability.

The database design follows **normalization principles**, ensuring minimal redundancy and maintaining data integrity through the use of primary and foreign keys.

Overall, the ERD provides a clear blueprint of the database structure and ensures that all analytical operations are supported by a well-defined and scalable relational model.

7. DATA CLEANING AND PREPARATION

Before performing analysis, the dataset underwent several preprocessing steps to ensure data accuracy, consistency, and reliability. Since the dataset consisted of multiple interconnected tables, careful cleaning and standardization were required to maintain integrity across relationships.

7.1 Data Cleaning Steps

The following steps were performed to prepare the dataset for analysis:

- Handling Missing Values: Missing or null values in certain fields were identified and handled appropriately, either by filtering out incomplete records or retaining them where they did not affect the analysis.
- Standardization of Date and Time Formats: Timestamp columns such as order purchase date, delivery date, and estimated delivery date were standardized to ensure consistency and enable time-based analysis.
- Column Renaming and Formatting: Column names were cleaned and standardized to maintain uniformity and improve query readability.
- Creation of Derived Metrics: New variables were created to support analysis, such as delivery time, calculated as the difference between order purchase date and delivery date.
- Data Type Validation: Data types were verified and corrected where necessary to ensure compatibility with SQL operations.

7.2 Challenges Addressed

During the cleaning process, several challenges were encountered:

- Presence of missing or incomplete records in certain tables
- Inconsistencies in formatting across different datasets
- Complexity arising from multiple interconnected tables
- Ensuring consistency while joining data from different sources

7.3 Outcome of Data Preparation

The data cleaning process resulted in a structured and reliable dataset suitable for analysis.

- Improved data consistency and accuracy
- Enhanced compatibility across tables for joins and queries
- Creation of meaningful metrics for deeper insights
- Increased reliability of analytical results

8. SQL IMPLEMENTATION

The SQL implementation phase focused on extracting meaningful insights from the structured relational database. A series of queries were developed to analyse key aspects of the dataset, including sales performance, customer behaviour, seller contribution, and delivery efficiency.

A combination of fundamental and advanced SQL techniques was applied to enable comprehensive analysis across multiple interconnected tables.

8.1 Query Techniques Used

- **Joins:** Joins were extensively used to integrate data from multiple tables such as orders, customers, payments, and order items. This enabled unified analysis across different entities and ensured that insights were derived from a consolidated dataset.

- **Aggregation Functions:** Functions such as SUM(), COUNT(), and AVG() were used to compute key business metrics, including total revenue, number of orders, and average delivery time.

- **Grouping and Filtering:** GROUP BY and HAVING clauses were applied to segment data and filter aggregated results. This was particularly useful for identifying top-performing sellers, high-revenue categories, and regional trends.

- **Subqueries:** Subqueries were used to perform nested analysis and compare values against overall dataset averages, enabling deeper insights into customer and seller performance.

- **CTEs and Window Functions:** Common Table Expressions (CTEs) and window functions were utilized for advanced analysis, such as ranking sellers and evaluating performance across partitions of data.

8.2 Query Design Approach

Queries were designed in a structured and incremental manner to ensure clarity and accuracy. Complex problems were broken down into smaller logical steps, improving readability and making the queries easier to debug and maintain.

Additionally, attention was given to query efficiency by minimizing redundant operations and ensuring proper use of joins and aggregations.

8.3 Integration with Analysis

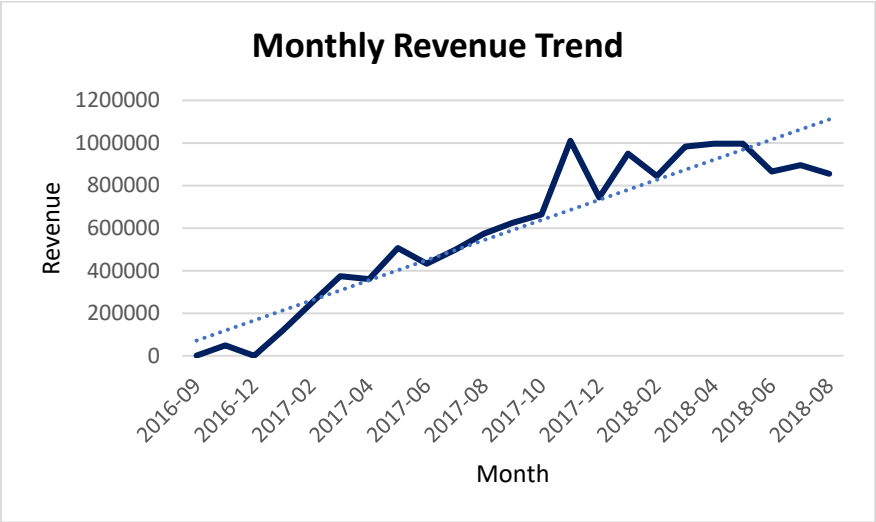
The SQL queries formed the foundation of the analytical process and directly supported all major sections of the project, including sales, customer, seller, and delivery analysis. The outputs generated from these queries were exported and used for visualization in Microsoft Excel, enabling clearer interpretation of trends and patterns.

9. DATA ANALYSIS

9.1 Sales Analysis

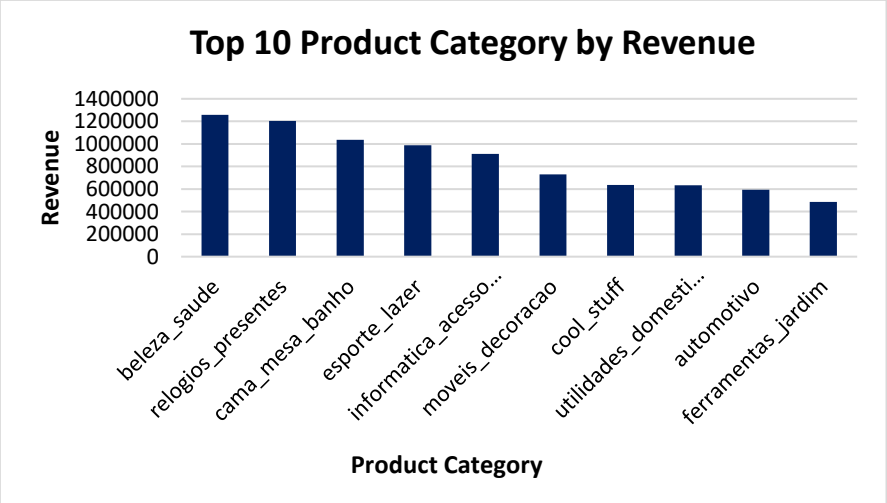
Sales analysis focuses on understanding revenue trends and order patterns. The analysis revealed fluctuations in sales over time, indicating potential seasonality in customer demand.

Monthly Revenue Trend



The monthly revenue trend shows a consistent increase from 2016 to early 2018, indicating strong business growth and rising customer demand. Although minor fluctuations are observed in later months, the overall trend appears to be stable reflecting a growth phase followed by gradual stabilization in the business.

Top 10 Product Categories by Revenue

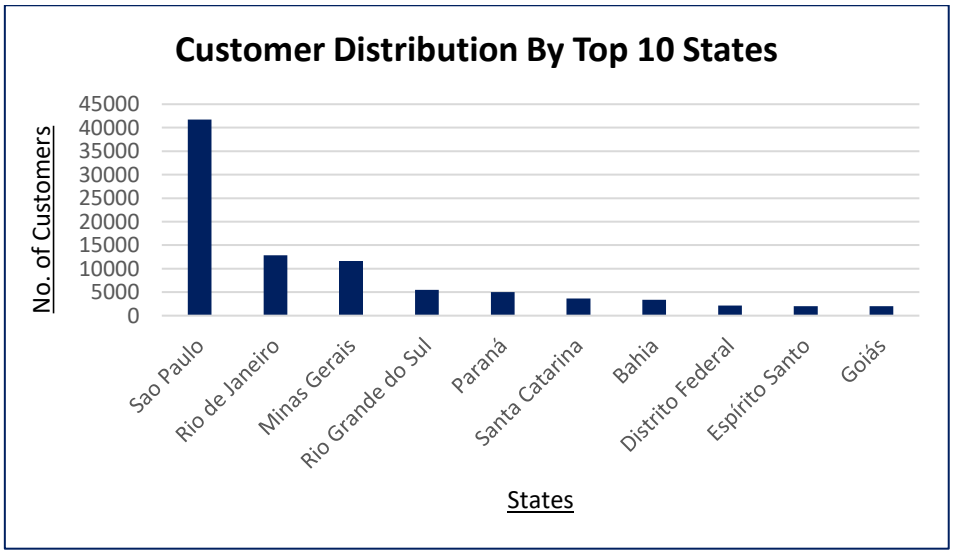


The analysis reveals that revenue is concentrated in a few key product categories, with categories such as beleza_saude and relogios_presentes generating the highest revenue. The remaining categories contribute comparatively less, indicating an uneven distribution of sales. This highlights the importance of top-performing categories in driving overall revenue.

9.2 Customer Analysis

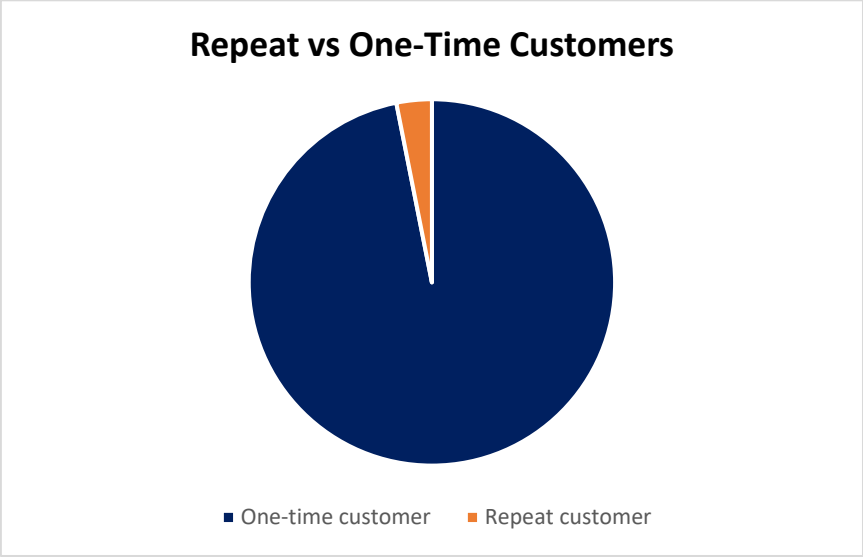
Customer analysis highlights purchasing behaviour and geographic distribution. It helps identify key customer segments and regions contributing to revenue.

Customer Distribution by State



The customer distribution is highly concentrated in a few states, with São Paulo having a significantly higher number of customers compared to others. Rio de Janeiro and Minas Gerais follow but at much lower levels. This indicates a strong regional concentration of customers in major urban areas.

Repeat vs One-Time Customers

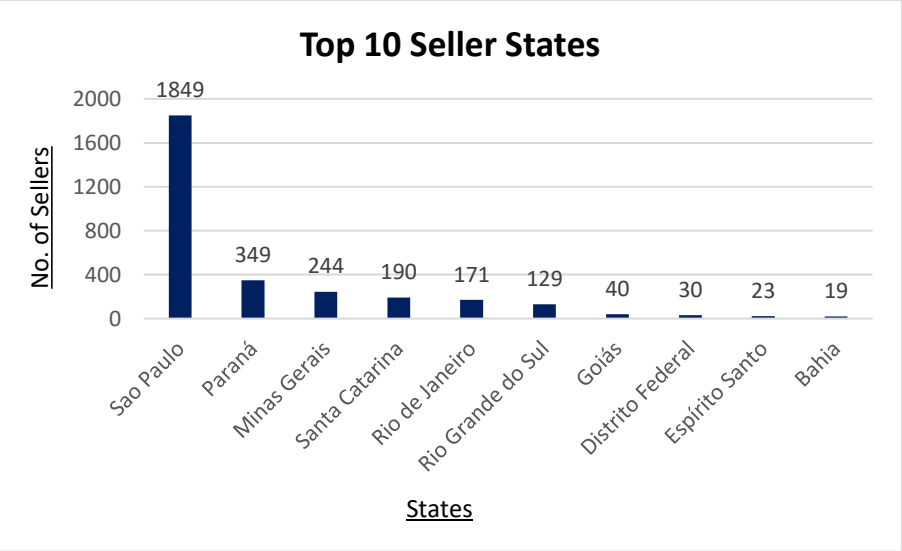


The majority of customers are one-time buyers, with repeat customers forming a very small proportion. This suggests low customer retention and highlights an opportunity to improve loyalty and repeat purchase behavior.

9.3 Seller Analysis

The analysis shows that a small group of sellers contributes significantly to overall revenue, indicating a concentration of performance.

Sellers by State

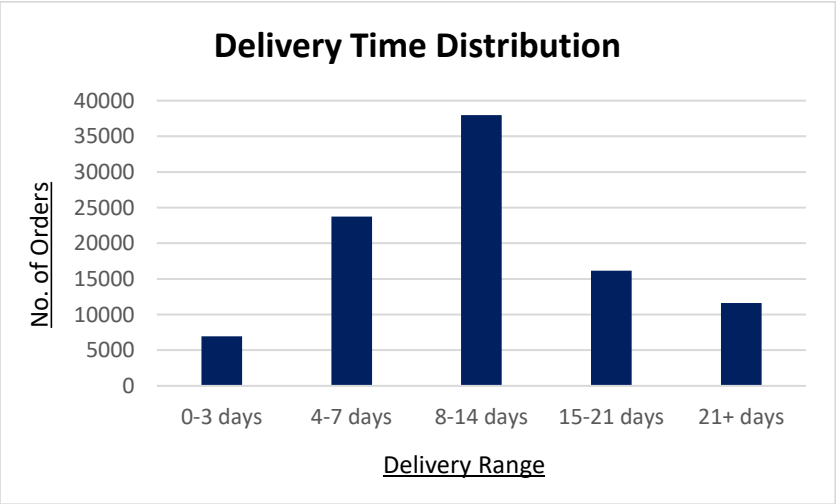


The distribution of sellers is uneven, with São Paulo dominating in terms of total sellers. Other states contribute significantly fewer sellers, indicating a geographic concentration of seller activity. This may impact regional availability and delivery efficiency.

9.4 Delivery and Logistics Analysis

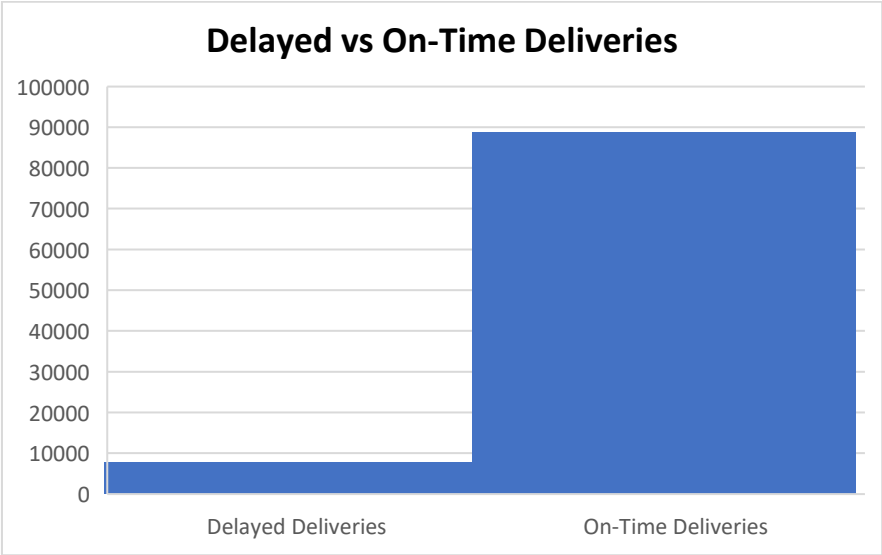
Delivery analysis reveals inefficiencies in logistics, particularly in certain regions where delays are more frequent.

Delivery Time Distribution



Most orders are delivered within the 8–14 day range, making it the most common delivery timeframe. Shorter delivery durations are less frequent, while longer delivery times still account for a noticeable share. This indicates moderate delivery efficiency with scope for improvement.

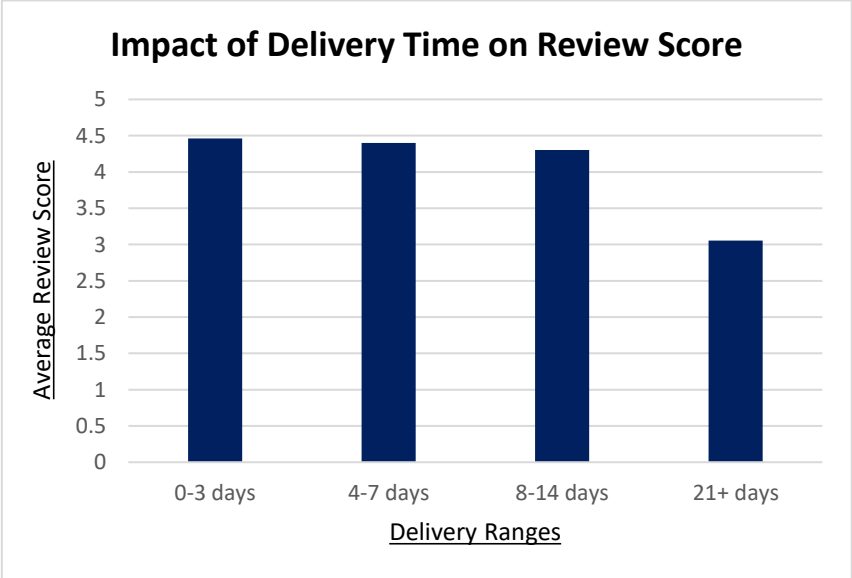
Delayed vs On-Time Deliveries



A large majority of deliveries are completed on time, while delayed deliveries represent a smaller proportion. This suggests that the logistics system is generally reliable, although delays still occur and may impact customer satisfaction.

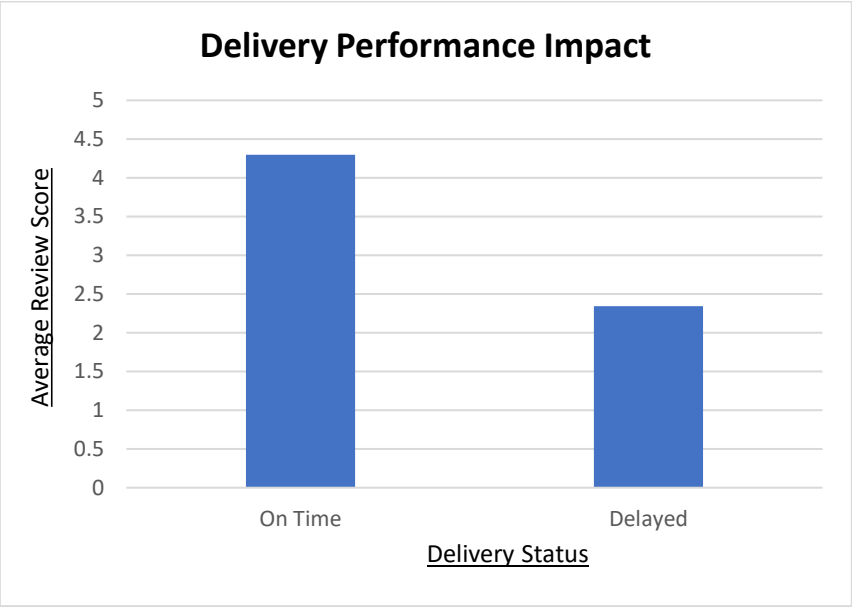
9.5 Advanced analytical insights

Impact of Delivery Time on Review Score



Review scores tend to decrease as delivery time increases, indicating a negative relationship between delivery duration and customer satisfaction. Faster deliveries receive higher ratings, highlighting the importance of efficient logistics.

Delivery Performance Impact



On-time deliveries are associated with significantly higher average review scores compared to delayed deliveries. This reinforces the importance of timely delivery in maintaining customer satisfaction and overall service quality.

10. KEY FINDINGS

The analysis of the e-commerce dataset reveals several important patterns across sales performance, customer behaviour, seller distribution, and delivery efficiency.

- The revenue trend shows consistent growth from 2016 to early 2018, followed by slight fluctuations and stabilization in later months, indicating a maturing sales pattern rather than continuous rapid growth.
- A significant portion of revenue is concentrated in a few product categories such as beleza_saude, relogios_presentes, and cama_mesa_banho, indicating strong customer demand in lifestyle and household segments.
- Customer distribution is highly concentrated in São Paulo, which alone accounts for over 41,000 customers, followed by Rio de Janeiro and Minas Gerais. This suggests that major urban centers dominate platform usage.
- Similarly, seller distribution is also heavily concentrated in São Paulo (1849 sellers), highlighting a regional clustering of supply.
- The average delivery time is approximately 12.5 days, with most orders falling within the 8-14 day range, indicating moderate logistical efficiency.
- A large majority of deliveries (~88,649 orders) are completed on time, while delayed deliveries (~7,827) represent a smaller but still significant proportion.
- The advanced analysis clearly shows that on-time deliveries receive significantly higher review scores (~4.5) compared to delayed deliveries (~2.5), emphasizing the critical role of delivery performance in customer satisfaction.
- The dataset also indicates that one-time customers (93,099) significantly outnumber repeat customers (2,997), suggesting low customer retention.

11. BUSINESS INSIGHTS AND RECOMMENDATIONS

The findings provide actionable insights that can help improve business performance and customer satisfaction.

- Since revenue is highly concentrated in a few categories, the business should **focus on expanding and promoting high-performing categories** such as beauty, home decor, and lifestyle products. At the same time, underperforming categories should be evaluated for improvement or repositioning.
- The heavy concentration of customers and sellers in São Paulo suggests an opportunity to **expand into underrepresented regions** through targeted marketing campaigns and seller onboarding initiatives.
- The low number of repeat customers highlights a major concern in customer retention. The company should introduce **loyalty programs, personalized recommendations, and targeted email marketing** to encourage repeat purchases.
- Delivery performance has a direct impact on customer satisfaction. Therefore, the company should **optimize logistics operations**, especially in regions with higher delays, by improving warehouse distribution and partnering with more efficient delivery providers.
- Since delayed deliveries result in significantly lower review scores, the business should implement **real-time tracking systems and proactive customer communication** to manage expectations and reduce dissatisfaction.
- The average delivery time of 12.5 days suggests room for improvement. Reducing delivery time even by a few days could significantly enhance customer experience and retention.

12. PERFORMANCE OPTIMIZATION

Based on the analysis, several performance optimization strategies can be identified to improve operational efficiency and business outcomes.

- **Logistics Optimization:** Since most deliveries fall within the 8-14 day range, efforts should be made to shift this distribution towards the 4-7 day range by optimizing supply chain networks and warehouse locations.
- **Regional Expansion Strategy:** With both customers and sellers concentrated in a few states, the company should optimize regional performance by identifying **high-potential but underutilized markets**.
- **Inventory Management:** High-performing product categories should be prioritized in inventory planning to avoid stockouts and ensure consistent availability.
- **Seller Performance Monitoring:** Since seller distribution is uneven, implementing **performance tracking systems for sellers** can help identify top-performing sellers and improve overall marketplace quality.
- **Customer Retention Strategies:** Given the high proportion of one-time customers, implementing **retention-focused strategies** such as discounts for repeat purchases, subscription models, or membership benefits can improve lifetime value.
- **Delivery Optimization:** Reducing delayed deliveries should be a key focus area. Predictive analytics can be used to identify potential delays and take corrective actions in advance.

13. LIMITATIONS OF THE ANALYSIS

While the analysis provides valuable insights, it is important to acknowledge certain limitations that may affect the interpretation of results.

- The analysis is limited to historical data and does not account for **external factors** such as economic conditions, seasonal demand, or market competition.
- Customer behaviour analysis is constrained due to the absence of detailed demographic information such as age, income, or preferences.
- The dataset does not include detailed operational data such as warehouse locations or delivery routes, limiting deeper logistics optimization analysis.
- The use of aggregated data may mask **individual-level variations**, especially in customer spending behaviour.
- The analysis assumes that all recorded transactions are accurate and does not account for potential data entry errors or missing values.

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14. CONCLUSION

This project successfully demonstrates the application of SQL and data analysis techniques to extract meaningful insights from an e-commerce dataset. By designing a structured relational database, performing comprehensive analysis, and visualizing key metrics, the project provides a clear understanding of business performance.

The findings highlight the importance of customer concentration, product category performance, and delivery efficiency in driving overall success. The strong relationship between delivery performance and customer satisfaction underscores the need for efficient logistics systems.

Additionally, the analysis reveals a critical challenge in customer retention, with a significantly higher proportion of one-time customers. Addressing this issue through targeted strategies can significantly enhance long-term profitability.

Overall, the project showcases the ability to transform raw data into actionable insights, demonstrating both technical proficiency in SQL and analytical thinking in business decision-making.

15.FUTURE IMPROVEMENTS

There are several opportunities to enhance the scope and depth of this project in the future.

- Incorporating **predictive analytics** can help forecast sales trends and customer behaviour, enabling proactive decision-making.
- Adding **customer segmentation techniques** can provide deeper insights into different customer groups and their purchasing patterns.
- Integration with **real-time data sources** can improve the accuracy and relevance of analysis.
- Expanding the analysis to include **profit margins and cost data** would provide a more comprehensive view of business performance.
- Implementing **dashboard tools such as Power BI or Tableau** can enhance data visualization and make insights more interactive.
- Further analysis of **delivery routes and logistics networks** can improve operational efficiency.
- Including **A/B testing and experimentation frameworks** can help evaluate the effectiveness of business strategies.

APPENDIX

Appendix A: SQL Queries and Outputs

1. Monthly Revenue Trend

Shows how revenue changes over time to identify sales trends.

```sql
-- Monthly Revenue Trend
-- (Display how total revenue varies month-by-month to understand the platform's sales growth and revenue fluctuations over time)
SELECT
DATE_FORMAT(o.order_purchase_timestamp,'%Y-%m') AS month,
ROUND(SUM(oi.price),2) AS revenue
FROM orders o
JOIN order_items oi
ON o.order_id = oi.order_id
GROUP BY month
ORDER BY month;
```

	A	B	C
1	month	revenue	
2	2016-09	267.36	
3	2016-10	49507.66	
4	2016-12	10.9	
5	2017-01	120312.9	
6	2017-02	247303	
7	2017-03	374344.3	
8	2017-04	359927.2	
9	2017-05	506071.1	
10	2017-06	433038.6	
11	2017-07	498031.5	
12	2017-08	573971.7	
13	2017-09	624401.7	
14	2017-10	664219.4	
15	2017-11	1010271	
16	2017-12	743914.2	
17	2018-01	950030.4	
18	2018-02	844178.7	
19	2018-03	983213.4	
20	2018-04	996647.8	
21	2018-05	996517.7	
22	2018-06	865124.3	
23	2018-07	895507.2	
24	2018-08	854686.3	
25			

2. Top 10 Product Categories by Revenue

Shows the top 10 products categories generating the highest revenue

```
-- Top 10 Product Category by Revenue
-- (Identifies the product categories generating the highest revenue)
    SELECT
    p.product_category_name,
    ROUND(SUM(oi.price),2) AS revenue
    FROM order_items oi
    JOIN products p
    ON oi.product_id = p.product_id
    GROUP BY p.product_category_name
    ORDER BY revenue DESC
    LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

product_category_name	revenue
beleza_saude	1258681.34
relogios_presentes	1205005.68
cama_mesa_banho	1036988.68
esporte_lazer	988048.97
informatica_acessorios	911954.32
moveis_decoracao	729762.49
cool_stuff	635290.85
utilidades_domesticas	632248.66
automotivo	592720.11
ferramentas_jardim	485256.46

3. Top 10 Sellers by Revenue

Identifies the top-performing sellers based on total revenue.

```sql
-- Top 10 Sellers by Revenue
-- (Identifies the top 10 sellers generating the highest revenue through product sales)
SELECT
seller_id,
ROUND(SUM(price),2) AS revenue
FROM order_items
GROUP BY seller_id
ORDER BY revenue DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

seller_id	revenue
4869f7a5dfa277a7dca6462dcf3b52b2	229472.63
53243585a1d6dc2643021fd1853d8905	222776.05
4a3ca9315b744ce9f8e9374361493884	200472.92
fa1c13f2614d7b5c4749cbc52fecda94	194042.03
7c67e1448b00f6e969d365cea6b010ab	187923.89
7e93a43ef30c4f03f38b393420bc753a	176431.87
da8622b14eb17ae2831f4ac5b9dab84a	160236.57
7a67c85e85bb2ce8582c35f2203ad736	141745.53
1025f0e2d44d7041d6cf58b6550e0bfa	138968.55
955fee9216a65b617aa5c0531780ce60	135171.70

4. Impact of Delivery Time on Review Score

Analyses how delivery time affects customer review ratings.

```sql
-- Impact of Delivery Time on Review Score
-- (Analyzes how delivery duration influences customer review scores and satisfaction levels)
SELECT
AVG(DATEDIFF(o.order_delivered_customer_date, o.order_purchase_timestamp)) AS avg_delivery_days,
AVG(r.review_score) AS avg_review_score
FROM orders o
JOIN order_reviews r
ON o.order_id = r.order_id
WHERE o.order_delivered_customer_date IS NOT NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_delivery_days	avg_review_score
12.4528	4.1576

5. **Repeat vs One-Time Customers**

Compares the number of repeat customers with one-time buyers.

```sql
-- Number of Repeat Customers vs One-Time Customers
-- (Compares customers who placed multiple orders with those who made only a single purchase to evaluate customer retention)
SELECT
CASE
WHEN order_count = 1 THEN 'One-time customer'
ELSE 'Repeat customer'
END AS customer_type,
COUNT(*) AS total_customers
FROM
(
SELECT
c.customer_unique_id,
COUNT(o.order_id) AS order_count
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_unique_id
) AS customer_orders
GROUP BY customer_type;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_type	total_customers
One-time customer	93099
Repeat customer	2997

6. **Delayed Deliveries**

Identifies orders that were delivered later than expected.

```sql
-- Delayed Deliveries
-- (Measures how many orders were delivered ater the estimated delivery date)
SELECT
COUNT(*) AS delayed_orders
FROM orders
WHERE order_delivered_customer_date > order_estimated_delivery_date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

delayed_orders
7827

7. **Average Delivery Time**

Calculates the average time taken to deliver orders.

```sql
-- Average Delivery Time
-- (Calculates the average number of days taken to deliver orders)
    SELECT
    AVG(DATEDIFF(order_delivered_customer_date, order_purchase_timestamp)) AS avg_delivery_days
    FROM orders
    WHERE order_delivered_customer_date IS NOT NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_delivery_days
12.4973

Appendix B: GitHub Repository Link

https://github.com/avneet0778/E-Commerce-Sales-and-Profitability-Warehouse-Project